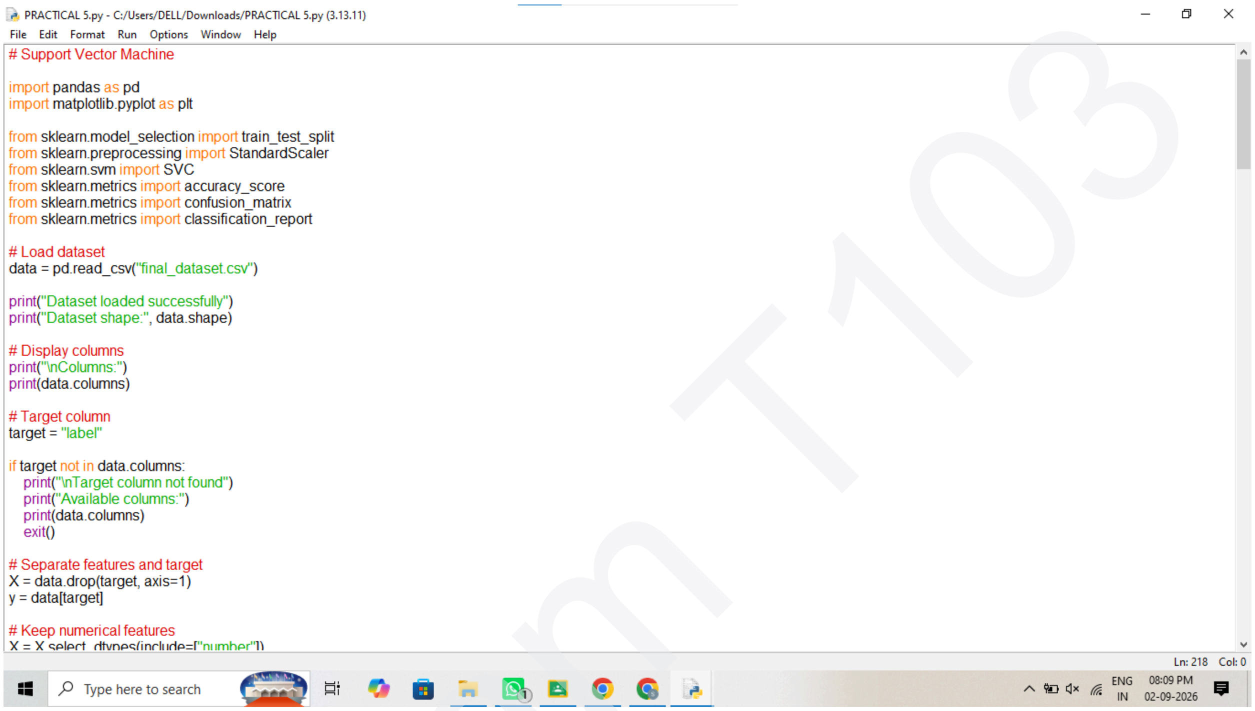
SHETH L.U.J & SIR M.V COLLEGE OF SCIENCE
SUBJECT : AI

Aim: Support Vector Machines (SVM)
Implement the SVM algorithm for binary classification.
Train an SVM model using a given dataset and optimize its parameters.
Evaluate the performance of the SVM model on test data and analyze the results.

Input



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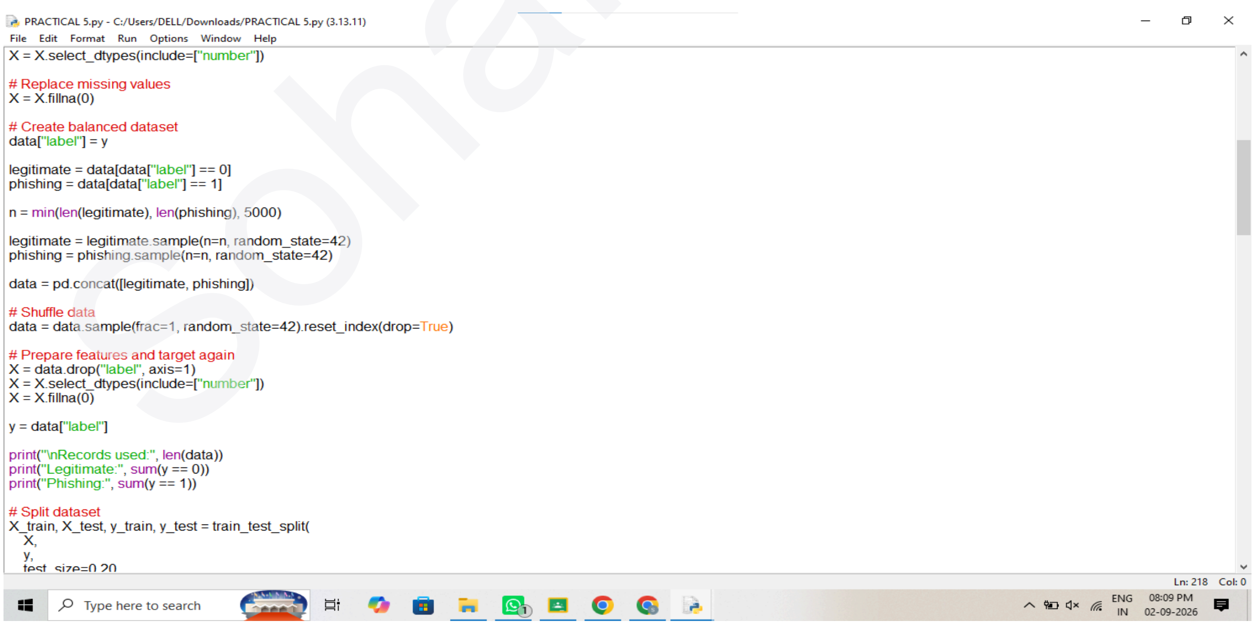
```
# Support Vector Machine

import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score
from sklearn.metrics import confusion_matrix
from sklearn.metrics import classification_report

# Load dataset
data = pd.read_csv("final_dataset.csv")

print("Dataset loaded successfully")
print("Dataset shape:", data.shape)

# Display columns
print("\nColumns:")
print(data.columns)

# Target column
target = "label"

if target not in data.columns:
    print("\nTarget column not found")
    print("Available columns:")
    print(data.columns)
    exit()

# Separate features and target
X = data.drop(target, axis=1)
y = data[target]

# Keep numerical features
X = X.select_dtypes(include=["number"])
```

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```
X = X.select_dtypes(include=["number"])

# Replace missing values
X = X.fillna(0)

# Create balanced dataset
data["label"] = y

legitimate = data[data["label"] == 0]
phishing = data[data["label"] == 1]

n = min(len(legitimate), len(phishing), 5000)

legitimate = legitimate.sample(n=n, random_state=42)
phishing = phishing.sample(n=n, random_state=42)

data = pd.concat([legitimate, phishing])

# Shuffle data
data = data.sample(frac=1, random_state=42).reset_index(drop=True)

# Prepare features and target again
X = data.drop("label", axis=1)
X = X.select_dtypes(include=["number"])
X = X.fillna(0)

y = data["label"]

print("\nRecords used:", len(data))
print("Legitimate:", sum(y == 0))
print("Phishing:", sum(y == 1))

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20
```


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```
    random_state=42
)

print("\nTraining records:", len(X_train))
print("Testing records:", len(X_test))

# Scale features
scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

# Parameter optimization
c_values = [0.1, 1, 10, 100]
accuracies = []

print("\nParameter Optimization")

for c in c_values:

    svm = SVC(
        kernel="rbf",
        C=c,
        gamma="scale"
    )

    svm.fit(X_train, y_train)

    prediction = svm.predict(X_test)

    accuracy = accuracy_score(y_test, prediction)

    accuracies.append(accuracy)

    print(
        "C =", c,
        "Accuracy =", round(accuracy * 100, 2), "%"
```

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```
    )

# Find best C value
best_index = accuracies.index(max(accuracies))
best_c = c_values[best_index]

print("\nBest C value:", best_c)
print(
    "Best Accuracy:",
    round(accuracies[best_index] * 100, 2),
    "%"
)

# Train final SVM
model = SVC(
    kernel="rbf",
    C=best_c,
    gamma="scale"
)

print("\nTraining final SVM...")

model.fit(X_train, y_train)

print("Training completed")

# Prediction
y_pred = model.predict(X_test)

# Final accuracy
final_accuracy = accuracy_score(y_test, y_pred)

print("\nFinal Accuracy:")
print(round(final_accuracy * 100, 2), "%")

# Confusion matrix
cm = confusion_matrix(y_test, y_pred)
```

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```python
cm = confusion_matrix(y_test, y_pred)

print("\nConfusion Matrix:")
print(cm)

# Classification report
print("\nClassification Report:")

print(
    classification_report(
        y_test,
        y_pred,
        target_names=["Legitimate", "Phishing"]
    )
)

# Parameter optimization graph
plt.figure(figsize=(7, 5))

plt.plot(
    c_values,
    accuracies,
    marker="o"
)

plt.title("SVM Parameter Optimization")

plt.xlabel("C Value")

plt.ylabel("Accuracy")

plt.xscale("log")

plt.grid()

plt.show()
```

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```python
plt.show()

# Confusion matrix graph
plt.figure(figsize=(6, 5))

plt.imshow(cm)

plt.title("SVM Confusion Matrix")

plt.xlabel("Predicted Class")

plt.ylabel("Actual Class")

plt.xticks(
    [0, 1],
    ["Legitimate", "Phishing"]
)

plt.yticks(
    [0, 1],
    ["Legitimate", "Phishing"]
)

for i in range(2):
    for j in range(2):
        plt.text(
            j,
            i,
            cm[i, j],
            ha="center",
            va="center"
        )

plt.colorbar()

plt.show()
```

Ln: 172 Col: 39

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Output

```
IDLE Shell 3.13.11
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Dataset loaded successfully
Dataset shape: (579920, 76)

Columns:
Index(['url', 'label', 'url_length', 'domain_length', 'hostname_length',
       'path_length', 'first_dir_length', 'tld_length', 'tld_length_domain',
       'url_depth', 'query_length', 'path_segments_count', 'num_digits',
       'num_letters', 'num_special_chars', 'num_dots', 'num_hyphens', 'num_at',
       'num_percent', 'num_equals', 'num_question', 'num_ampersand',
       'num_hash', 'num_underscore', 'num_special', 'num_slash', 'num_params',
       'entropy_url', 'entropy_hostname', 'entropy_domain', 'entropy_path',
       'query_entropy', 'ratio_digits', 'ratio_letters', 'ratio_special_chars',
       'uppercase_ratio', 'lowercase_ratio', 'is_ip_address', 'starts_with_ip',
       'is_suspicious_tld', 'uses_https', 'has_www', 'unusual_double_slash',
       'multiple_http', 'contains_port_number', 'path_has_encoded_chars',
       'query_has_base64', 'contains_login', 'contains_secure',
       'contains_verify', 'contains_account', 'contains_update',
       'contains_bank', 'contains_cloud', 'contains_brand', 'query_key_count',
       'query_value_length_avg', 'success', 'dns_resolves', 'has_mx_record',
       'has_txt_record', 'has_ns_record', 'ttl_value', 'ip_count',
       'cname_count', 'resolves_to_private_ip', 'whois_success',
       'domain_age_days', 'expiration_days', 'creation_year',
       'domain_is_recent', 'domain_registered_before_2020', 'registrar_valid',
       'name_servers_count', 'is_privacy_protected', 'whois_missing'],
      dtype='object')

Records used: 10000
Legitimate: 5000
Phishing: 5000

Training records: 8000
Testing records: 2000

Parameter Optimization
C = 0.1 Accuracy = 92.7 %
C = 1 Accuracy = 95.75 %
C = 10 Accuracy = 95.9 %
Ln: 67 Col: 0
```

```
IDLE Shell 3.13.11
File Edit Shell Debug Options Window Help
Records used: 10000
Legitimate: 5000
Phishing: 5000

Training records: 8000
Testing records: 2000

Parameter Optimization
C = 0.1 Accuracy = 92.7 %
C = 1 Accuracy = 95.75 %
C = 10 Accuracy = 95.9 %
C = 100 Accuracy = 95.6 %

Best C value: 10
Best Accuracy: 95.9 %

Training final SVM...
Training completed

Final Accuracy:
95.9 %

Confusion Matrix:
[[971  42]
 [ 40 947]]

Classification Report:
              precision    recall  f1-score   support

  Legitimate       0.96      0.96      0.96      1013
    Phishing       0.96      0.96      0.96       987

    accuracy                           0.96      2000
   macro avg       0.96      0.96      0.96      2000
weighted avg       0.96      0.96      0.96      2000

>>>
Ln: 67 Col: 0
```


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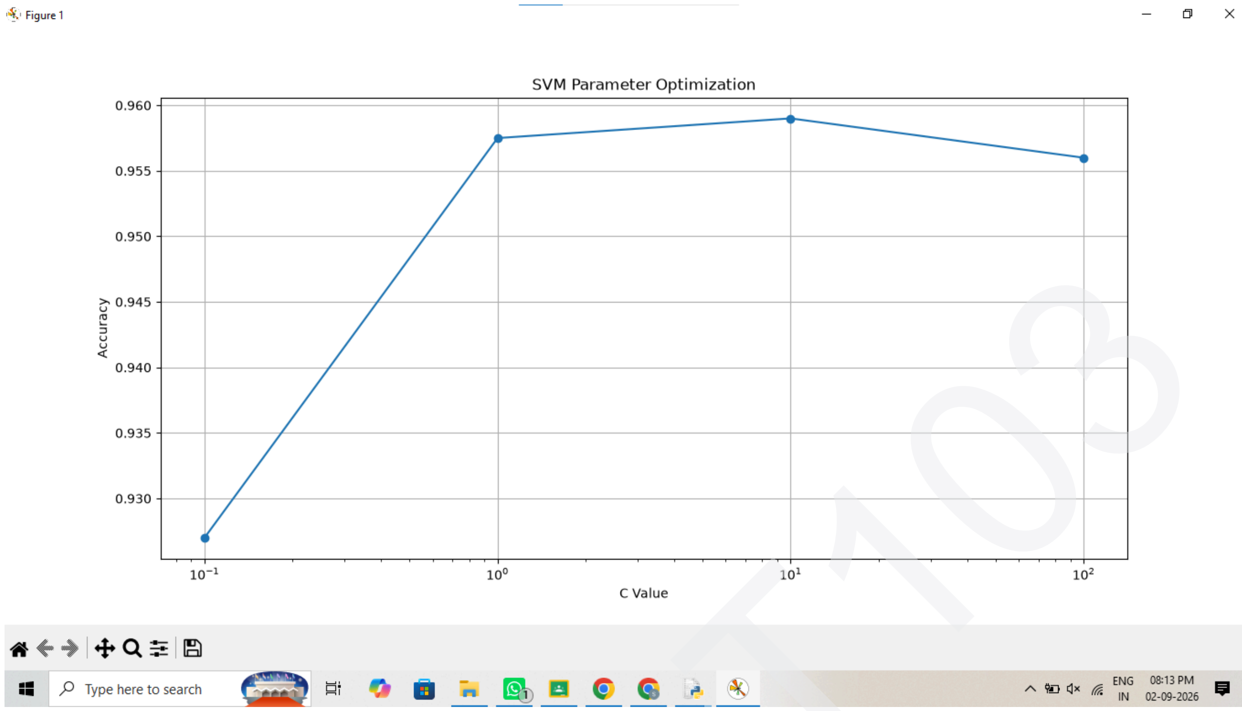

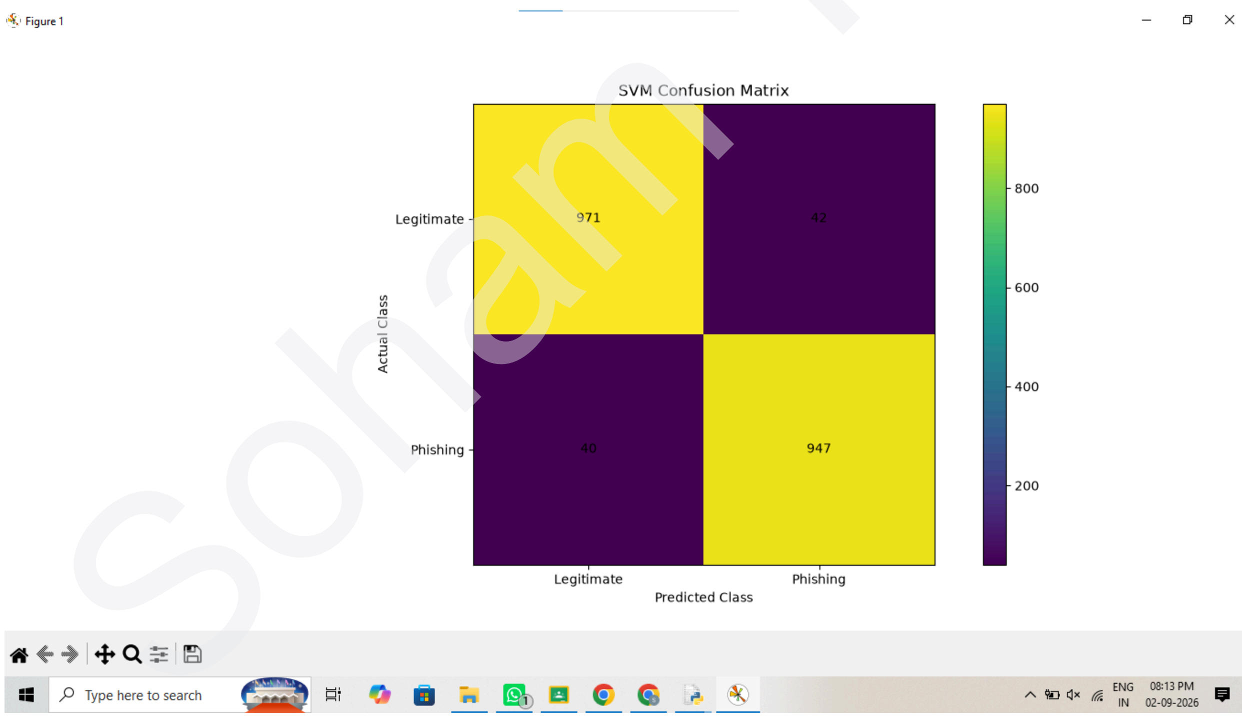

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